Introduction to Deep Learning for Text Processing

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Text Processing
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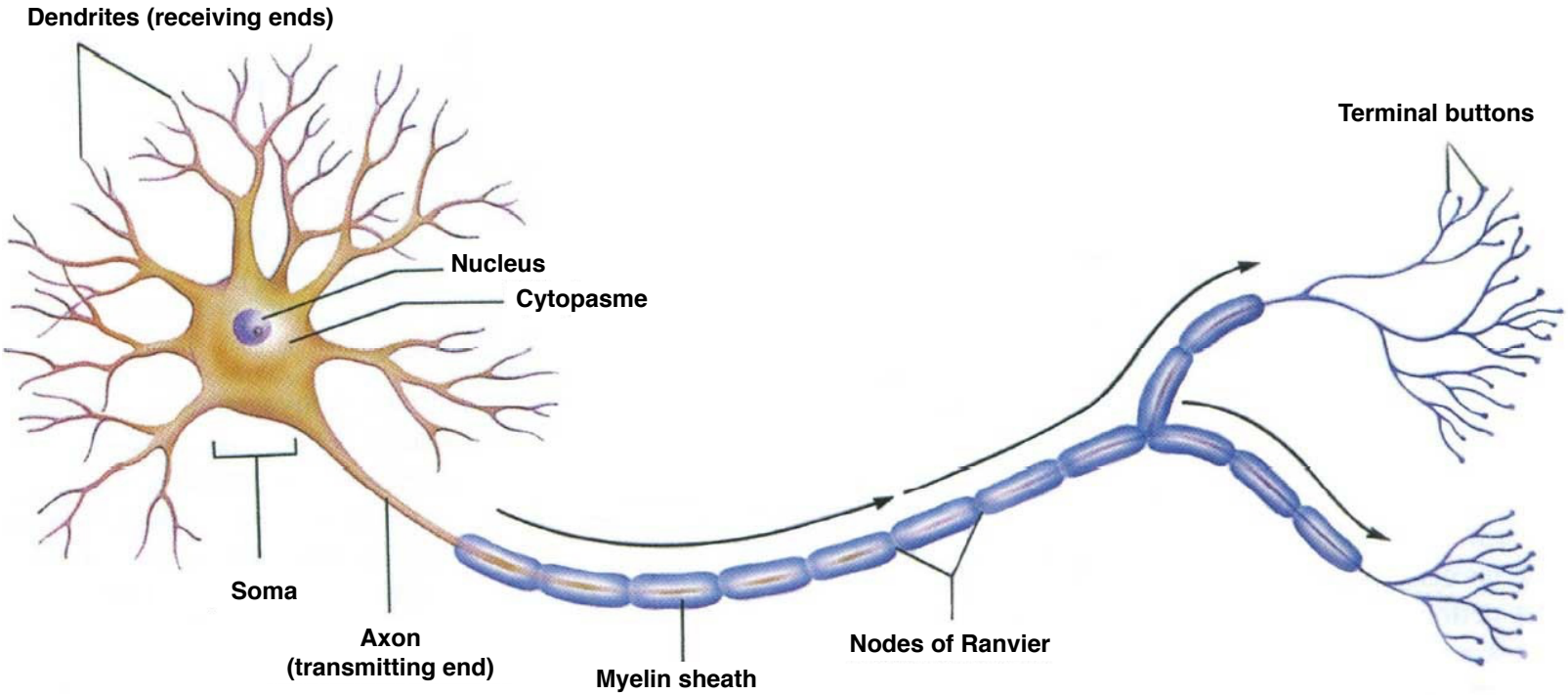
Learning Objectives

- Explain the skip-gram model for word embeddings
- Name some of the applications of word embeddings
- Name techniques for sentence representation

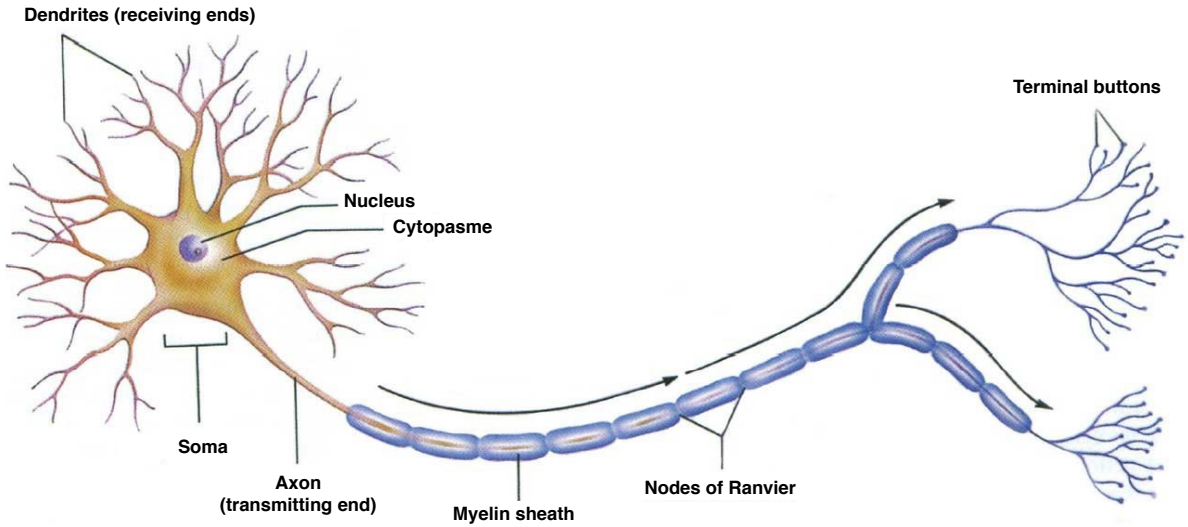
Text Processing: Deep Learning: Overview

- **Shortest introduction to Neural networks**
- Representing words
- Representing sentences
- Deep Learning for Sentiment Analysis
- Deep Learning for Information Extraction (NER)

Biological neuron / nerve cell

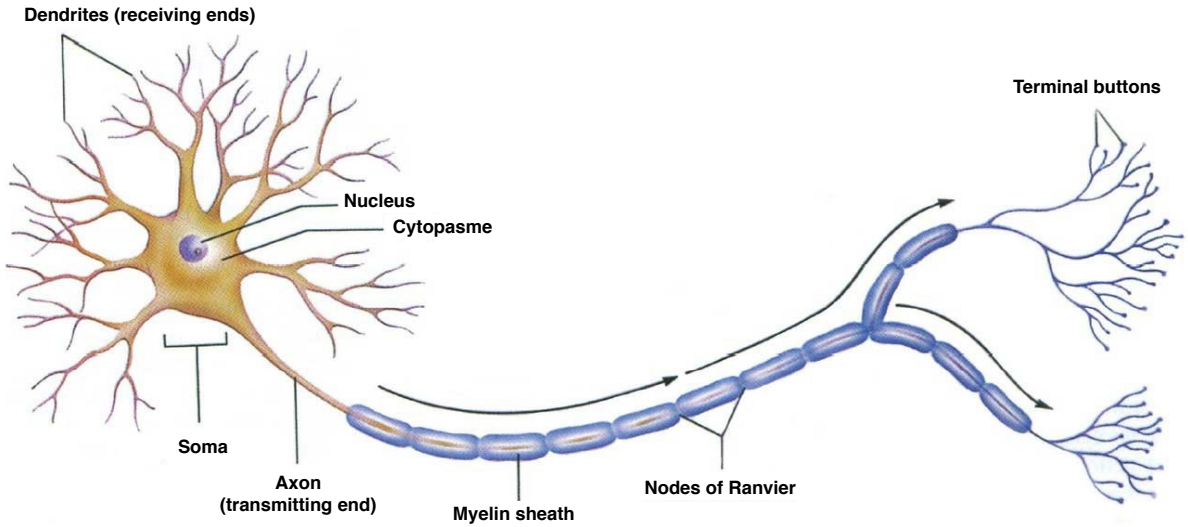


Hebb principle



Hebb: **"Neurons that fire together, wire together"**

Hebb principle



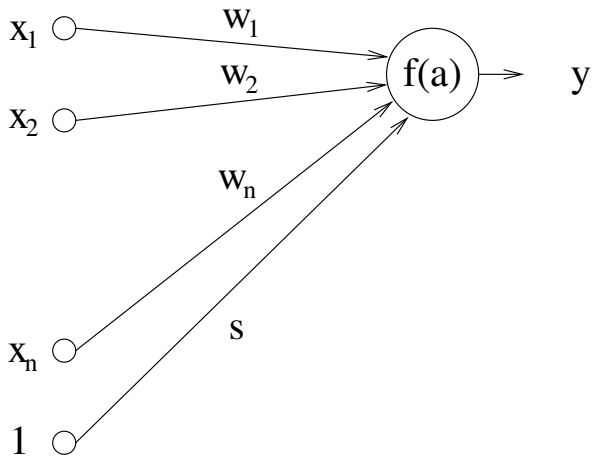
Hebb: **"Neurons that fire together, wire together"**

- Cells are active together $\rightarrow$ reinforce their connection
- Cells are not active together $\rightarrow$ diminish their connection

$\Rightarrow$ **local process** there is no global supervision

The perceptron

Perceptron: computing unit loosely inspired by the biological neuron

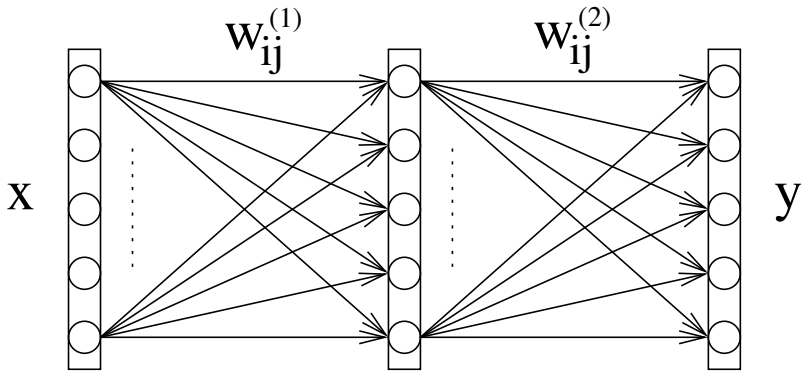


input:	$\mathbf{x} = \{x_i\}$
weights:	$\mathbf{w} = \{w_i\}$
threshold:	s
activity:	$a = \sum_i w_i x_i + s$
activation function:	$f =$ threshold function
output:	$\hat{y} = f(a)$

Training method: change the weights $\mathbf{w}$ if a training example $\mathbf{x}$ is misclassified as follows:

$$\mathbf{w}^{new} = \mathbf{w}^{cur} + \mathbf{x}.y \qquad \text{with} \qquad y \in \{+1, -1\}$$

Multilayer perceptron



$$
\begin{aligned}
y_i^2 &= f\left(\sum_j w_{ij}^1 \; x_j^1\right) \\
y_i^3 &= f\left(\sum_j w_{ij}^2 \; y_j^2\right) \\
&\vdots \\
y_i^c &= f\left(\sum_j w_{ij}^{c-1} \; y_j^{c-1}\right)
\end{aligned}
$$

$\Rightarrow$ **propagation** of the input **x** towards the output **y**

Multilayer perceptron: training

1. Normalise data
2. Initialise the weights $\mathbf{W}$
3. Repeat
   - Pick a **batch** of examples $(\mathbf{x}, \mathbf{c})$
   - **Forward** pass: propagate the batch $\mathbf{x}$ through the network $\rightarrow \mathbf{y}$
   - Calculate the error $E(\mathbf{y}, \mathbf{c})$
   - **Backward** pass: **backpropagation** $\rightarrow \nabla w_{ij}$
   - Update weights $w_{ij}^{new} = w_{ij}^{cur} - \lambda \frac{\partial E}{\partial w_{ij}}$
   - Eventually change the training meta-parameters (e.g. learning rate λ)

   until convergence

That's great, but where is the text?!?

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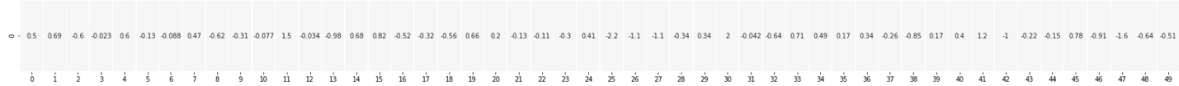
How to represent words?

The semantic properties of the words are encoded in the dimensions of the vector



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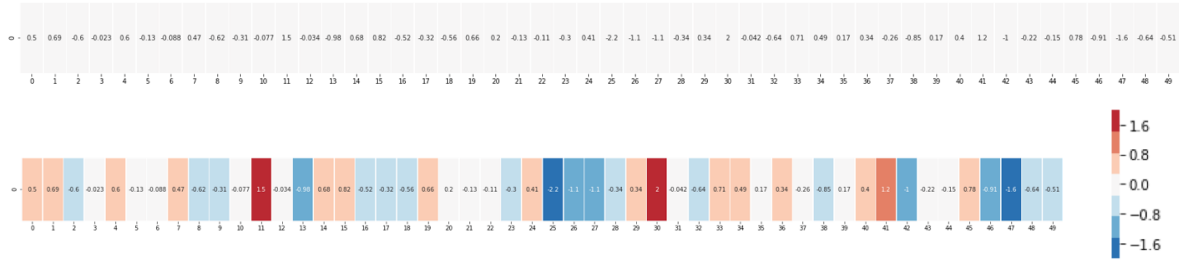

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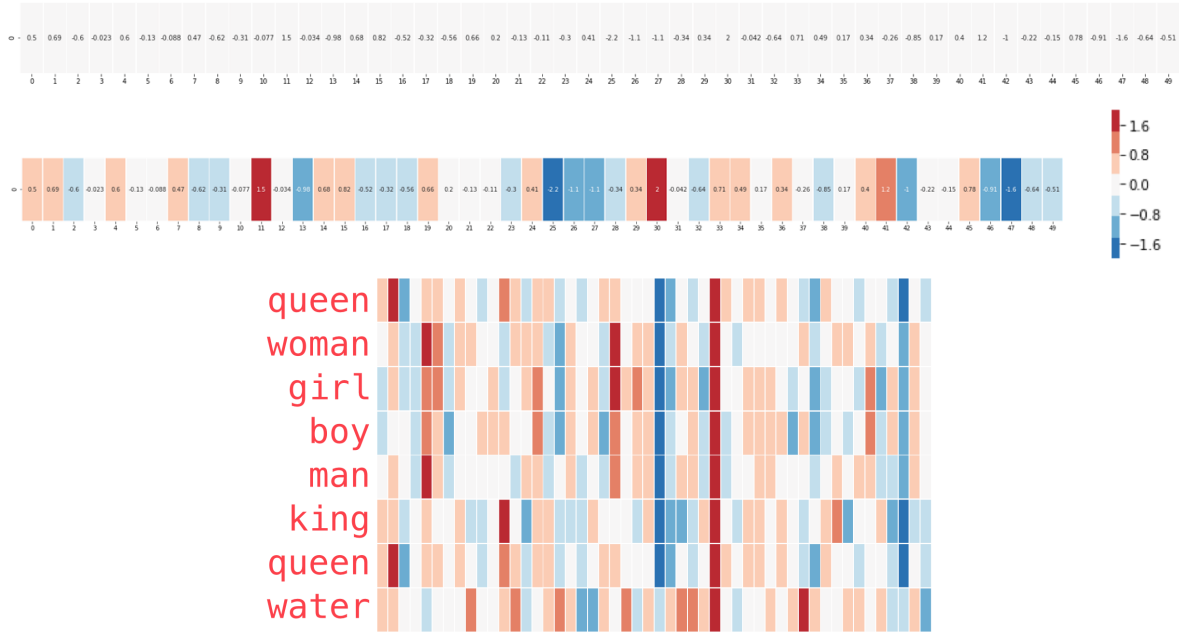

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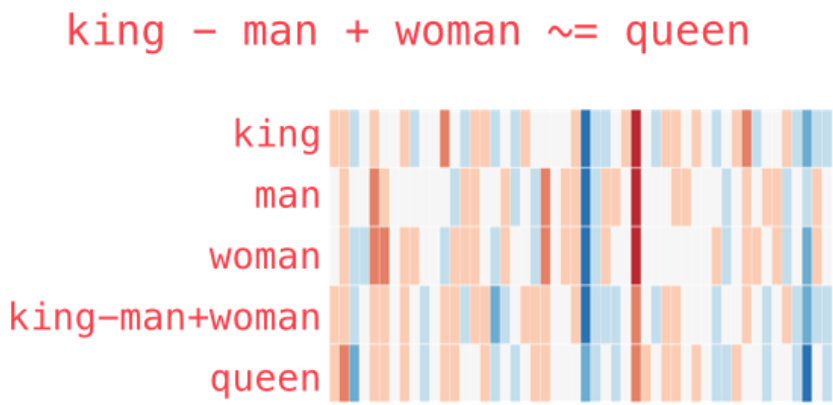

Can be learned in several ways:

- Extract handcrafted meaningful features
- Use a neural network!

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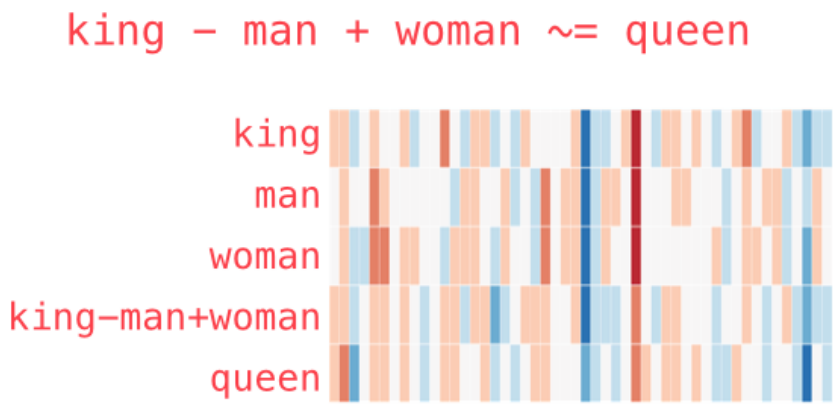

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- What would be the simplest way of obtaining vectors? $\Rightarrow$ so-called **1-hot vector**:
  - vector of size equal to **vocabulary size**
  - contains 0 everywhere except for a single 1 at a specific position

a	=	[1,0,0,0,0,0,...,0]
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...	...	...
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- Is that a good representation? ⇒ **NO!**
  - distance between any two words is the same for all word pairs
  - position of the "1" arbitrarily
  - → it is just a **coding**

Previous classes

- **Count-based distributional vectors**
  - Count number of times a **word co-occurs with other words**
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  - TFIDF, LSA, LDA, etc
- Problems:
  - Vector size **increases with vocabulary**
  - **High dimensionality**
- **SVD** $\rightarrow$ solve dimensionality issues, but
  - **Computational costs** is high $\rightarrow$ not feasible for millions of words or documents
  - **Difficult to add new words** to the model

Prediction-based distributional vectors

- **Neural network** algorithms
- Directly build **dense, low-dimensional** representations
- **Embeddings:** words are **embedded** in a low dimensional vector

Predict between **every word** and its **context words**

word2vec

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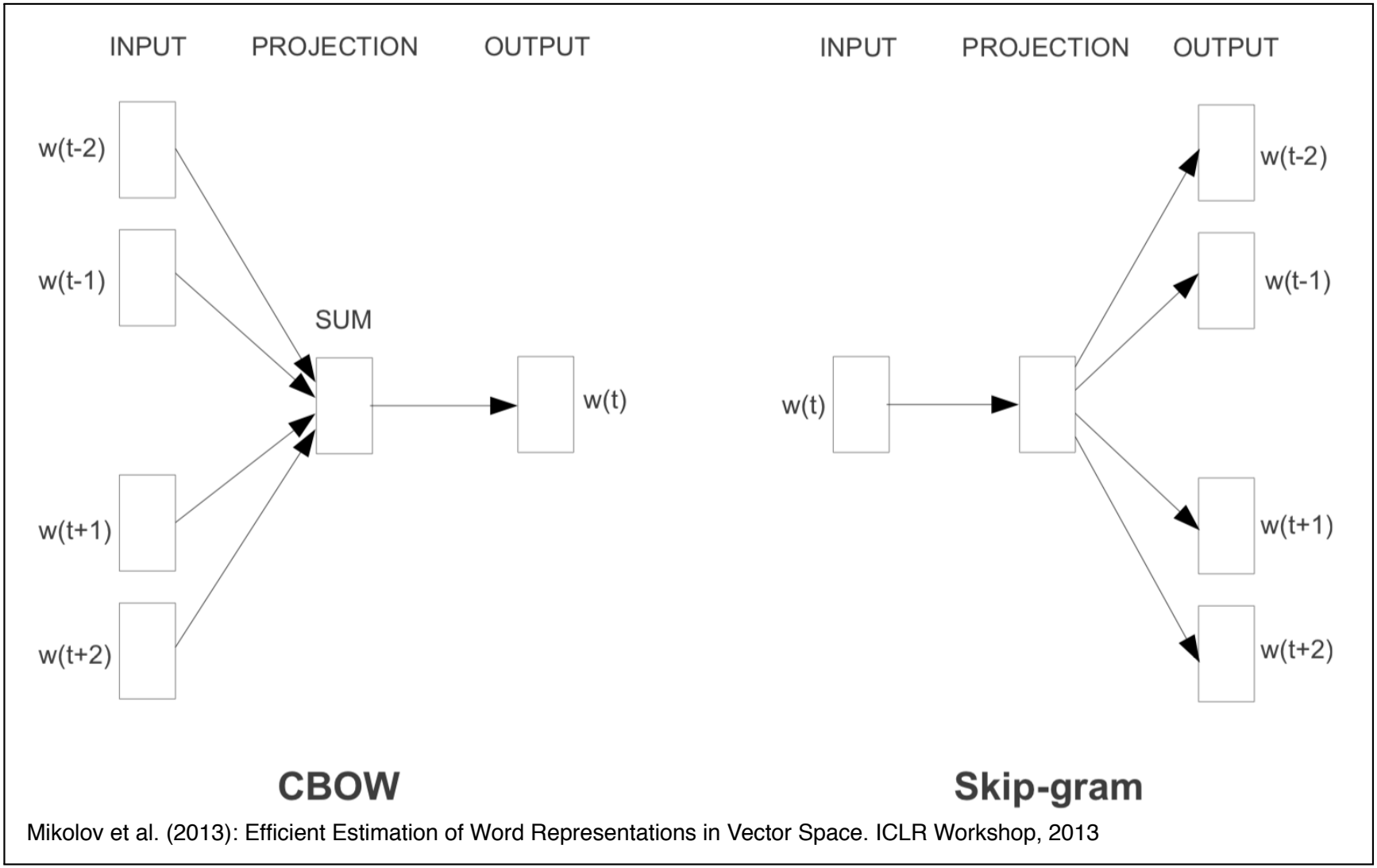

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word2vec

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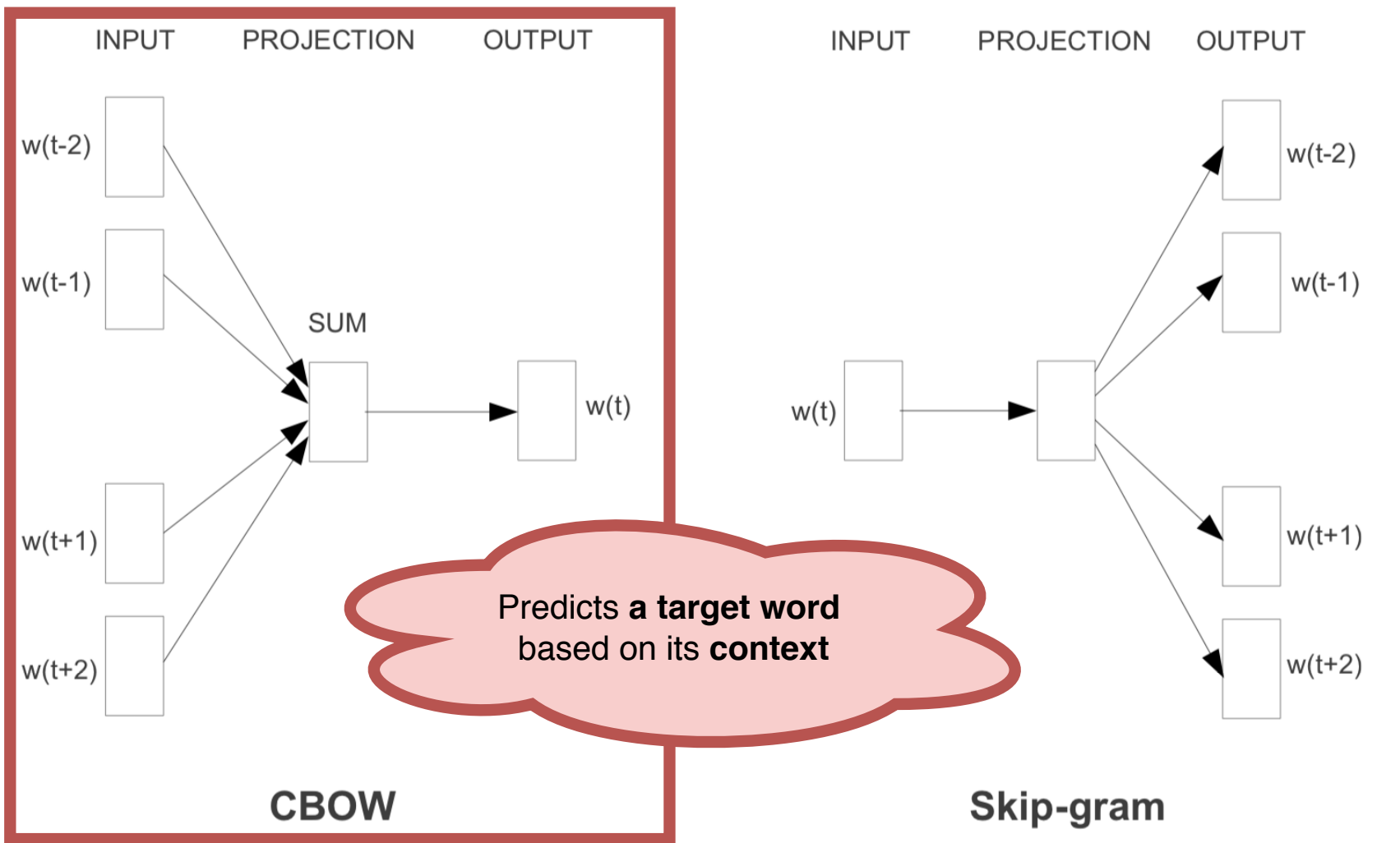

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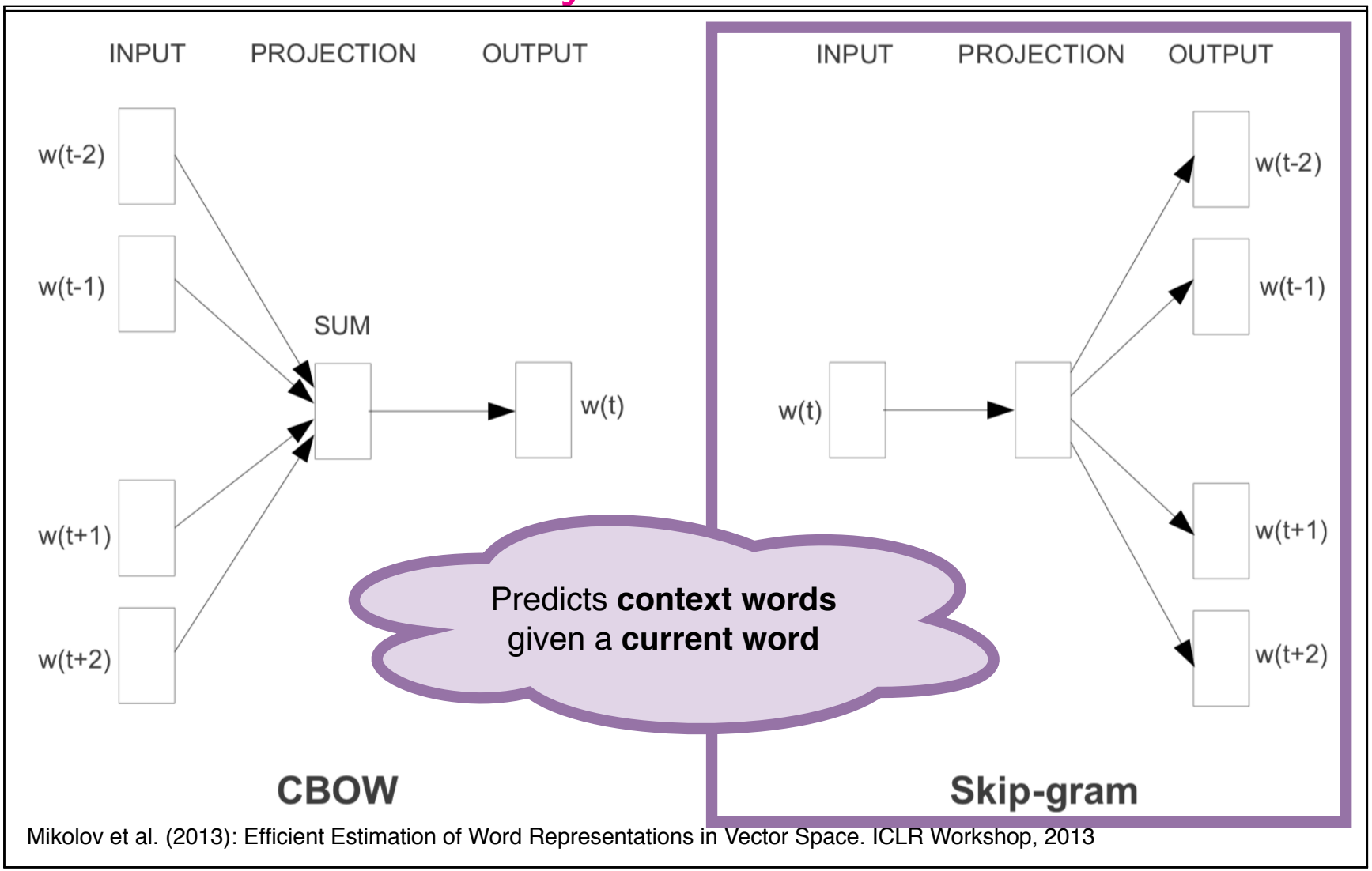

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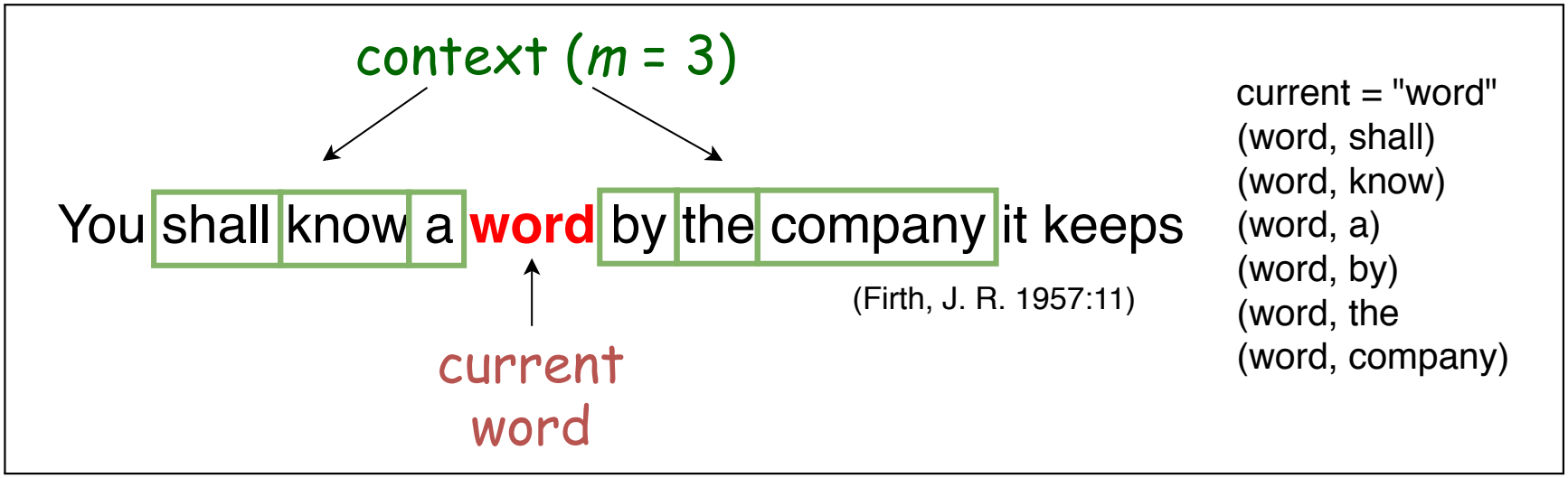
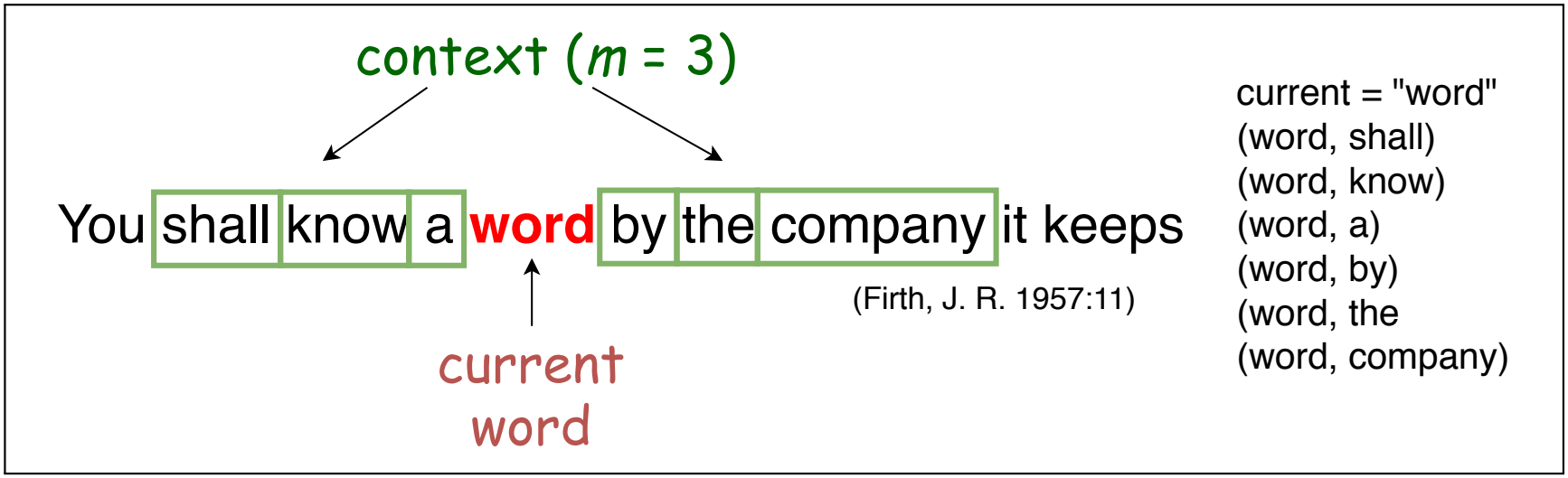
Skip-gram


context (m = 3)
You shall know a word by the company it keeps
(Firth, J. R. 1957:11)
current word
current = "word"
(word, shall)
(word, know)
(word, a)
(word, by)
(word, the
(word, company)

Skip-gram



Minimize:

$$J(\theta) = -\frac{1}{T}\sum_{t=1}^{T}\sum_{\substack{-m\leq j\leq m\\ i\neq 0}} \log p(w_{t+j}|w_t;\theta)$$

Skip-gram

$$p(w_{t+j}|w_t) = \frac{\exp({u_{w_{t+j}}}^\top v_{w_t})}{\sum_{w'=1}^{V} \exp({u_{w'}}^\top v_{w_t})}$$

- $\boldsymbol{u_w}$: vector representation of w as **context word**
- $\boldsymbol{v_w}$: vector representation of w as **current word**

Skip-gram

Softmax: from numbers to probability distributions

Exponentiate to make it positive

$$p(w_{t+j}|w_t) = \frac{\exp(u_{w_{t+j}}{}^\top v_{w_t})}{\sum_{w'=1}^{V} \exp(u_{w'}{}^\top v_{w_t})}$$

Normalised to give probability

- u_w: vector representation of w as **context word**
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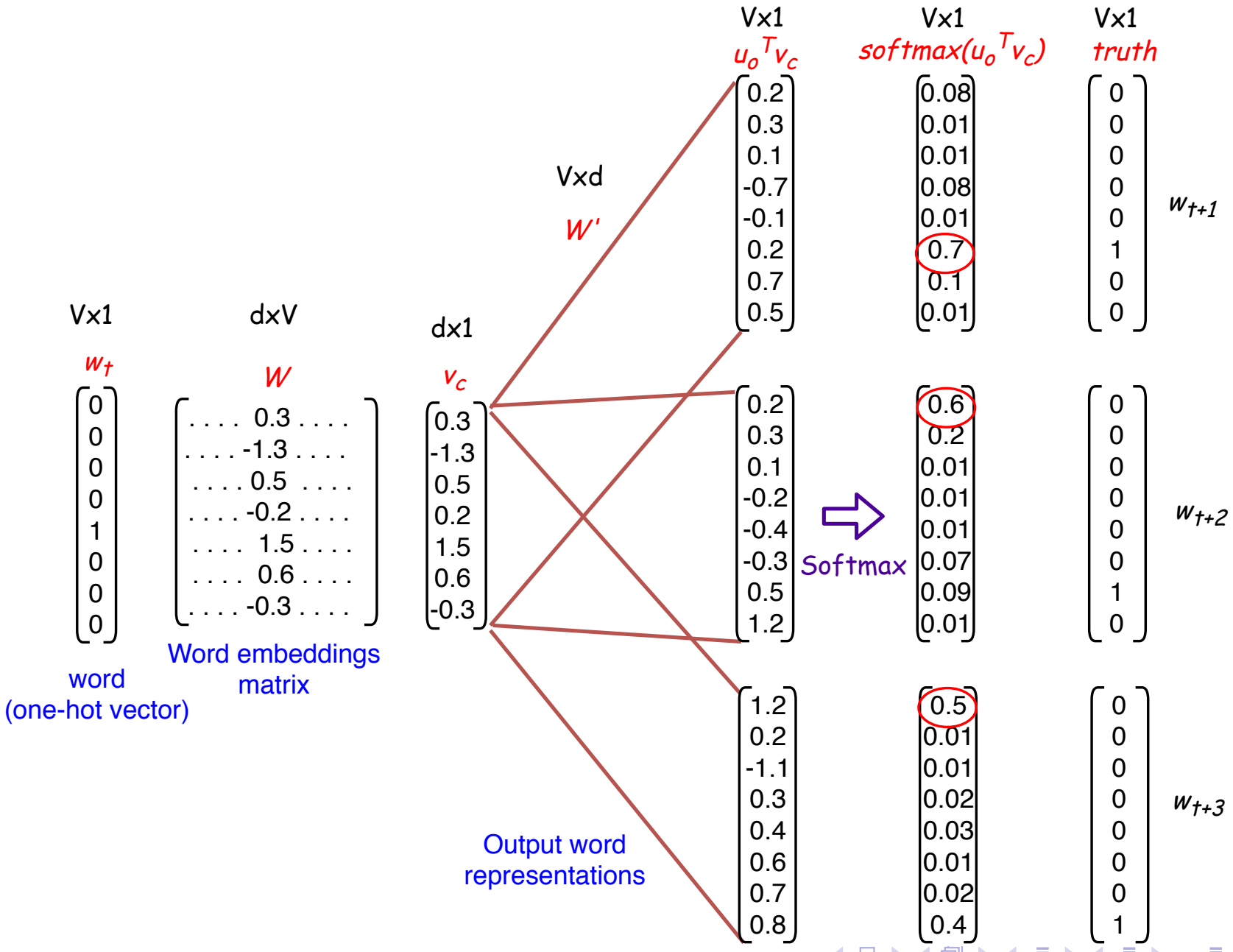
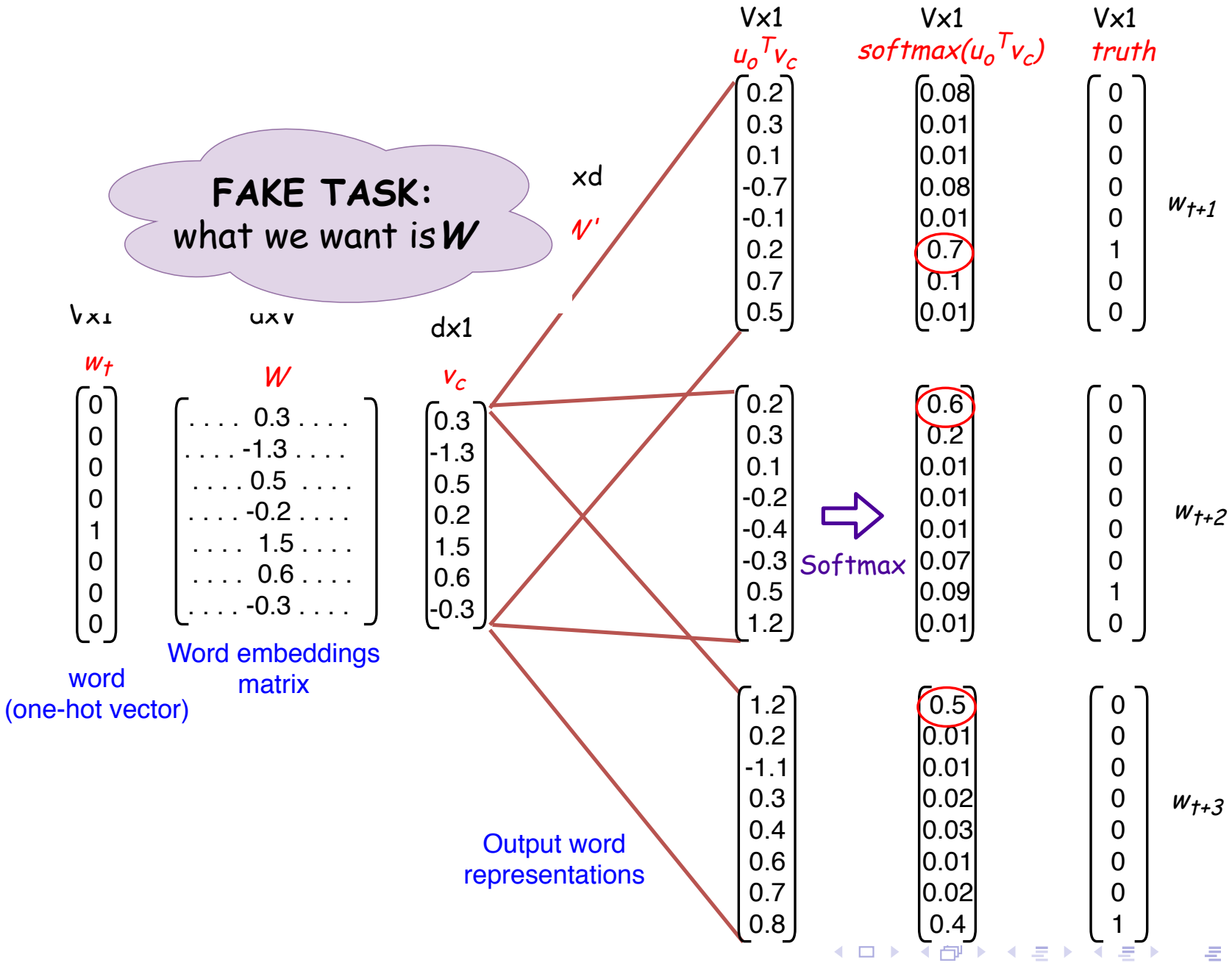
Skip-gram


Vx1
w_t
0
0
0
0
1
0
0
0
word
(one-hot vector)
dxV
W
. . . . 0.3 . . . .
. . . . -1.3 . . . .
. . . . 0.5 . . . .
. . . . -0.2 . . . .
. . . . 1.5 . . . .
. . . . 0.6 . . . .
. . . . -0.3 . . . .
Word embeddings
matrix
dx1
v_c
0.3
-1.3
0.5
0.2
1.5
0.6
-0.3
Vxd
W'
Output word
representations
Vx1
$u_o{}^T v_c$
0.2
0.3
0.1
-0.7
-0.1
0.2
0.7
0.5
0.2
0.3
0.1
-0.2
-0.4
-0.3
0.5
1.2
1.2
0.2
-1.1
0.3
0.4
0.6
0.7
0.8
Softmax
Vx1
$softmax(u_o{}^T v_c)$
0.08
0.01
0.01
0.08
0.01
0.7
0.1
0.01
0.6
0.2
0.01
0.01
0.01
0.07
0.09
0.01
0.5
0.01
0.01
0.02
0.03
0.01
0.02
0.4
Vx1
truth
0
0
0
0
0
1
0
0
w_{t+1}
0
0
0
0
0
0
1
0
w_{t+2}
0
0
0
0
0
0
0
1
w_{t+3}

Skip-gram


FAKE TASK:
what we want is W
Vx1
w_t
0
0
0
0
1
0
0
0
word
(one-hot vector)
dxV
W
. . . . 0.3 . . . .
. . . . -1.3 . . . .
. . . . 0.5 . . . .
. . . . -0.2 . . . .
. . . . 1.5 . . . .
. . . . 0.6 . . . .
. . . . -0.3 . . . .
Word embeddings
matrix
dx1
v_c
0.3
-1.3
0.5
0.2
1.5
0.6
-0.3
xd
W'
Output word
representations
Vx1
$u_o{}^T v_c$
0.2
0.3
0.1
-0.7
-0.1
0.2
0.7
0.5
0.2
0.3
0.1
-0.2
-0.4
-0.3
0.5
1.2
1.2
0.2
-1.1
0.3
0.4
0.6
0.7
0.8
Softmax
Vx1
$softmax(u_o{}^T v_c)$
0.08
0.01
0.01
0.08
0.01
0.7
0.1
0.01
0.6
0.2
0.01
0.01
0.01
0.07
0.09
0.01
0.5
0.01
0.01
0.02
0.03
0.01
0.02
0.4
Vx1
truth
0
0
0
0
0
1
0
0
w_{t+1}
0
0
0
0
0
0
1
0
w_{t+2}
0
0
0
0
0
0
0
1
w_{t+3}

Skip-gram

- Training the model: compute **all vector gradients**
  - θ is a long **vector of word vectors**
  - each word has **2 vectors**

$$\theta = \begin{bmatrix} v_{aardvak} \\ v_a \\ \dots \\ v_{zebra} \\ u_{aardvak} \\ u_a \\ \dots \\ u_{zebra} \end{bmatrix} \in \mathbb{R}^{2dV}$$

Skip-gram

$$J(\theta) = -\frac{1}{T}\sum_{t=1}^{T}\sum_{\substack{-m\leq j\leq m\\ i\neq 0}} \log p(w_{t+j}|w_t;\theta)$$

- How to **minimise** $J(\theta)$? $\rightarrow$ **(stochastic) gradient decent**
  - Calculate **gradients for each** v **and each** u on a window

- **Examples:** gradient of the log likelihood w.r.t. v_{w_t}

$$\frac{\partial \log p(w_{t+j}|w_t)}{\partial v_{w_t}} = u_{w_{t+j}} - \mathop{\mathbb{E}}_{x\sim p(x|w_t)}[u_x]$$

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Gradient calculations are left for you!

Christopher Manning: Natural Language Processing with Deep Learning @ Stanford University:
https://www.youtube.com/watch?v=ERibwqs9p38

Skip-gram

Problem: computing a single forward pass $\rightarrow$ **sum across the entire vocabulary**

- Example: for **300 dimensions** and **V = 10K words**
  - when training for each pair (w_t, w_{t+j})
  - **3M weights** in each the hidden layer and output layer

Solution: perform approximations!

- Hierarchical softmax
- **Negative sampling**

Negative sampling

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- Change the task: **binary logistic regression classifiers**

I want a glass of orange juice to go along with my cereal.

	x		y
	context	current word	target
	orange	**juice**	**1**
K	orange	king	0
	orange	book	0
	orange	the	0
	orange	of	0

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k	orange	of	0

Minimize:

Approximate $\mathbb{E}$ by drawing **sample of size** ***k*** from **noise distribution**

$$-\left(\log Q_\theta(y=1|w_{t+j}, w_t) + k \mathop{\mathbb{E}}_{\tilde{w}\sim P_{noise}} [\log Q_\theta(y=0|w_{t+j}, \tilde{w})]\right)$$

Probability of pair (w_{t+j}, w_t) to be **positive** calculated **according to** θ

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K	orange	the	0
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For $k = 4$
- update positive word ("juice")
- update 4 negative words
- output: 5 neurons, **1,500 weight values**
(0.05% of 3M)

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Why does it work?

- Let's assume that the word representations are **organised semantically**
- words w_1 and w_2 having similar meaning would be **close to each other** in this space

  $\rightarrow$ Consequently $\mathcal{F}(w_1) \approx \mathcal{F}(w_2)$

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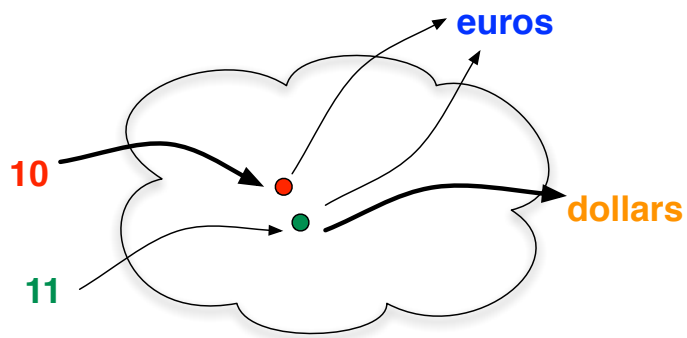
Language modelling:

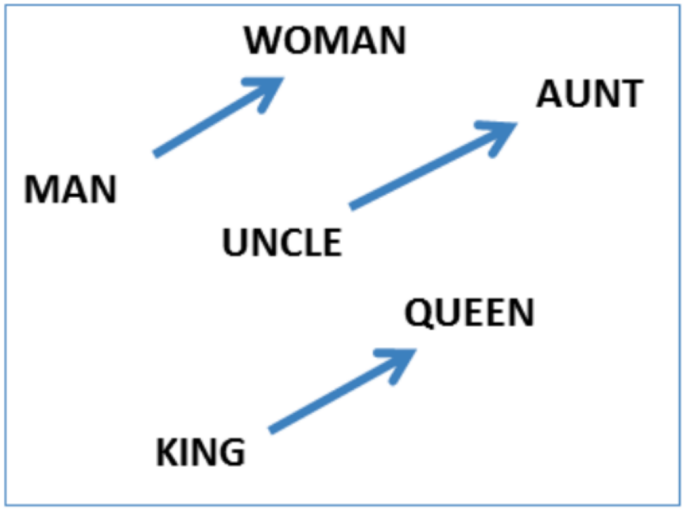
1. I have got **10** euros in my wallet
2. This item costs 11 euros
3. In the U.S. it is 11 **dollars** !



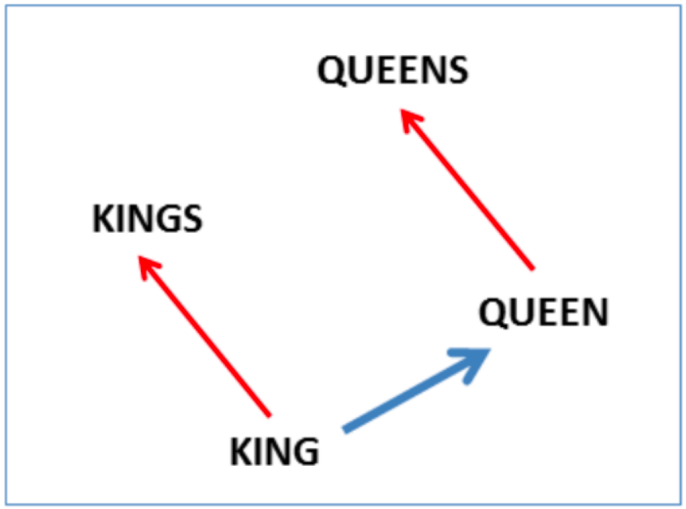
$\Rightarrow$ What is the probability that **10** is followed by **dollars**?

What can we do with word2vec?

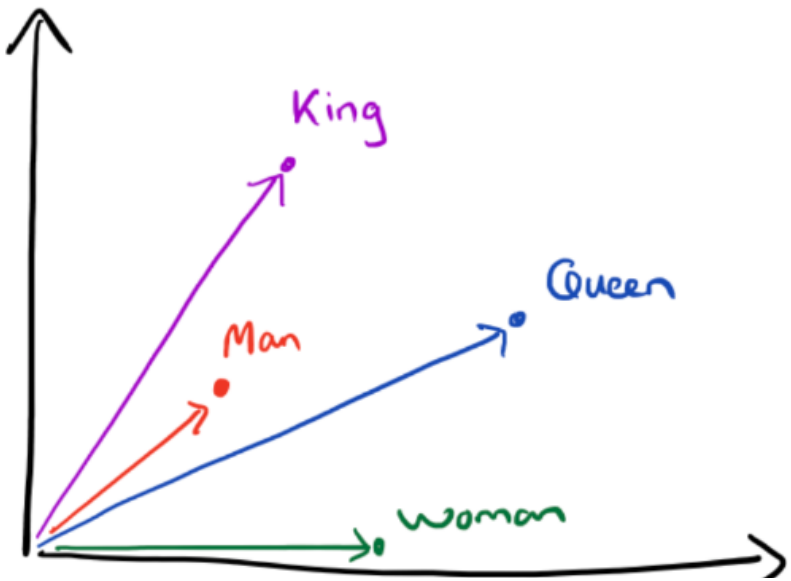
Relationships

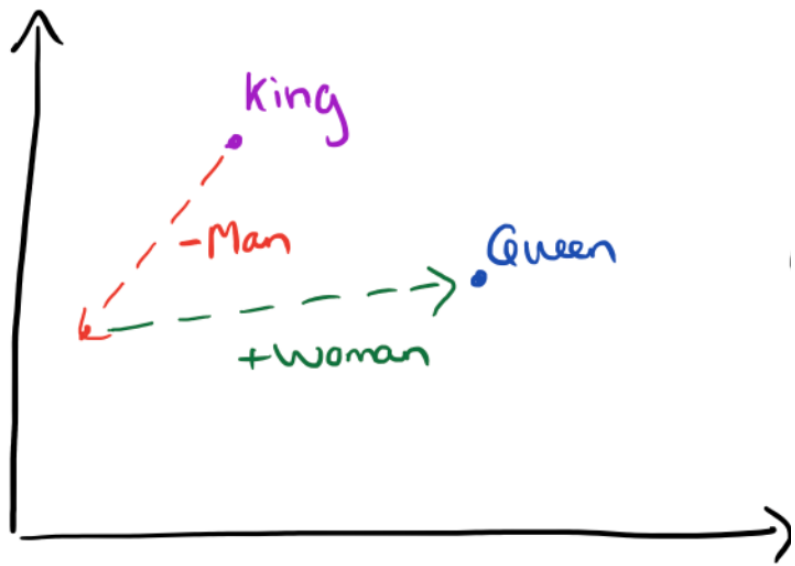


Plural



General semantics: king - man + woman = queen

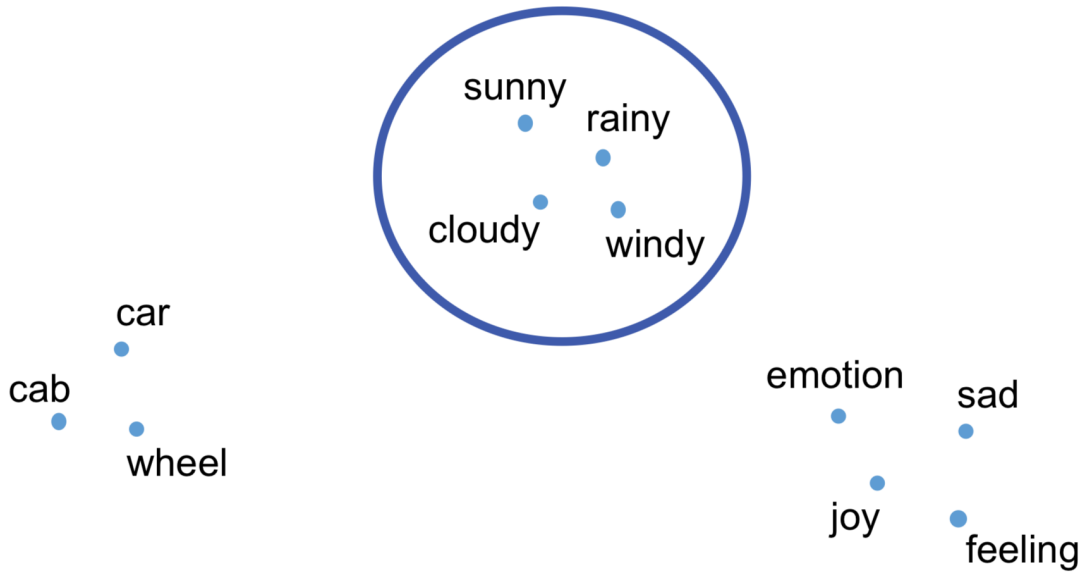




https://blog.acolyer.org/2016/04/21/the-amazing-power-of-word-vectors/

What can we do with word2vec?

Good to find **some relationship** between words



Kai-Wei Chang @ University of Virginia: http://www.cs.virginia.edu/~kc2wc/teaching/NLP16/slides/07-WordEmbedding.pdf

Text Processing: Deep Learning: Overview

- Shortest introduction to Neural networks
- Representing words
- **Representing sentences**
- Deep Learning for Sentiment Analysis
- Deep Learning for Information Extraction (NER)

How to represent sentences?

Sentence = sequence of word $\Rightarrow$ we need an **encoder**
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- **Recurrent neural network** (RNN)
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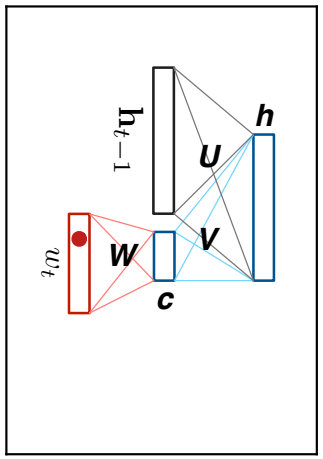

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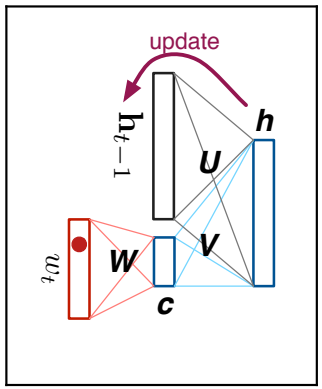

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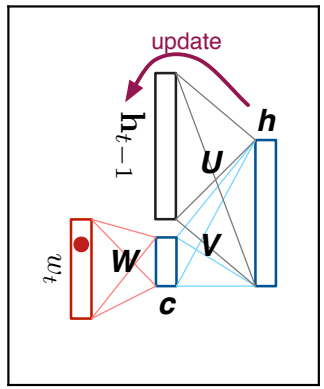

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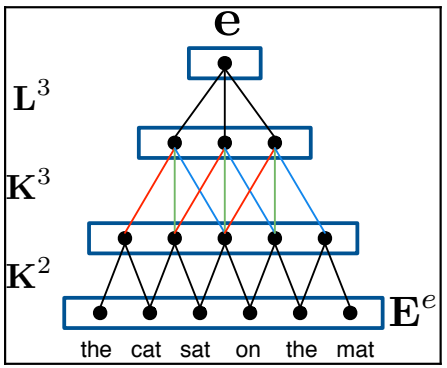

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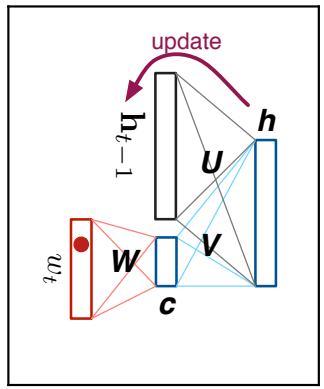

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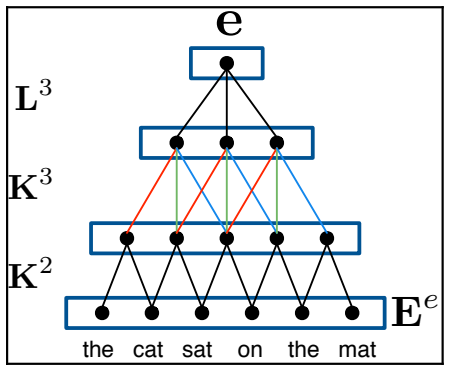
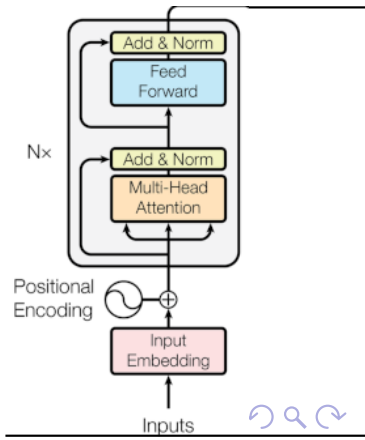


- Very recently **Transformers** = self-attention
  - representation = matrix (1 vector per word)
  - Must read:

    `http://jalammar.github.io/illustrated-transformer/`



How to classify sentences?

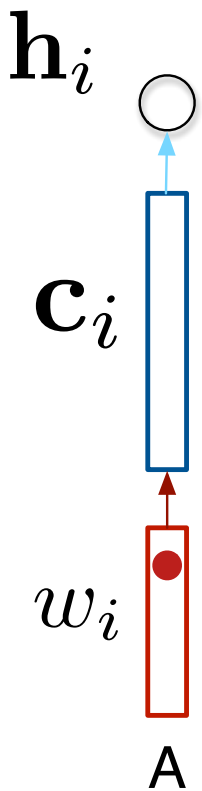
- The classifier is a neural network implementing a complex function $\mathcal{F}$
  - that operates in the **continuous space**
  - that maps input vectors to a **probability distribution** over the desired classes

1. Encode the sentence
   - get a vector
   - get a matrix (1 vector per word) $\rightarrow$ compress into 1 vector
     - **pooling** operation (usually mean or max)
     - concatenation
2. Non-linear classification layer(s) $\rightarrow$ get a vector of scores $\mathbf{z}$ (1 for each class)
3. Get a probability distribution by normalization $\rightarrow$ softmax

$$p(\mathbf{c} = j|\theta) = \frac{e^{\mathbf{z}_j}}{\sum_{k=1}^{\|V\|} e^{\mathbf{z}_k}}$$

Encoding a sentence with a (bi-)RNN

Sentence: "**A long time ago in a galaxy far , far away**"



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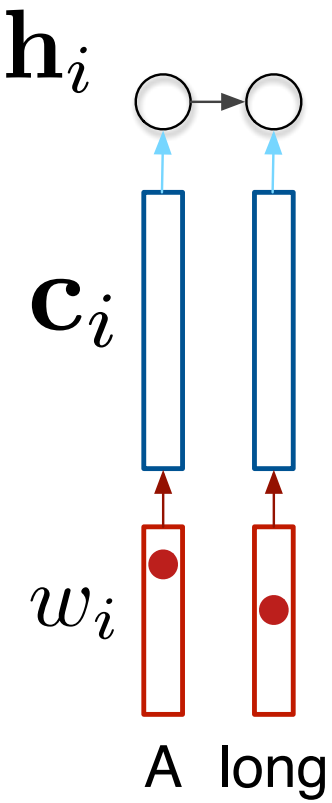

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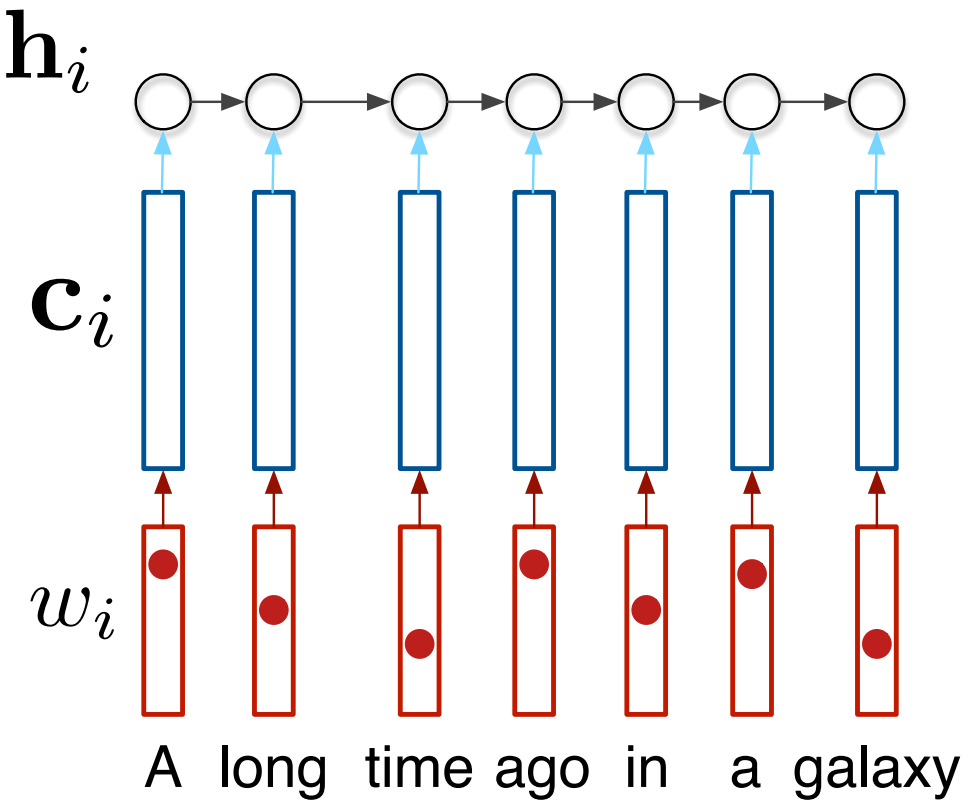

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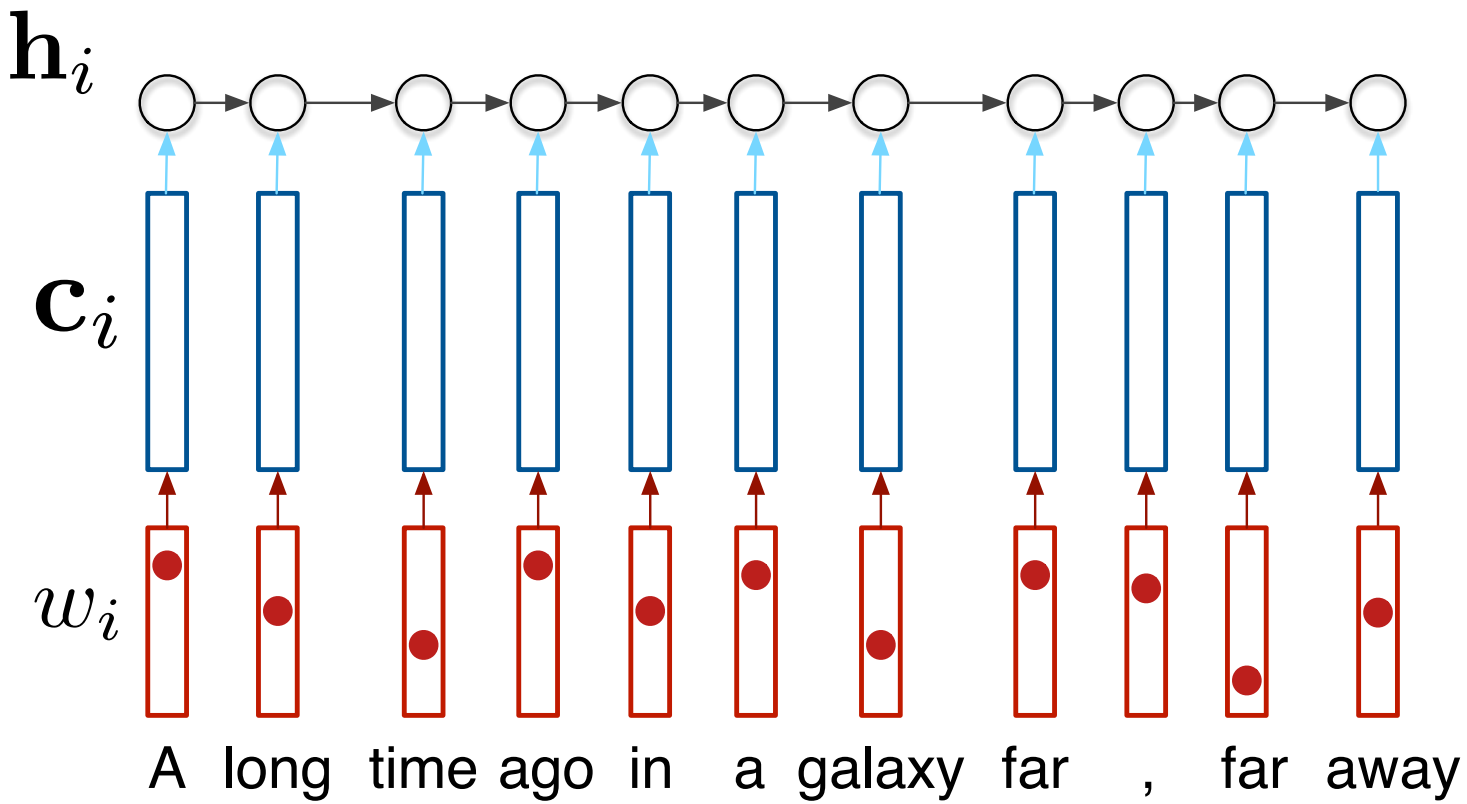

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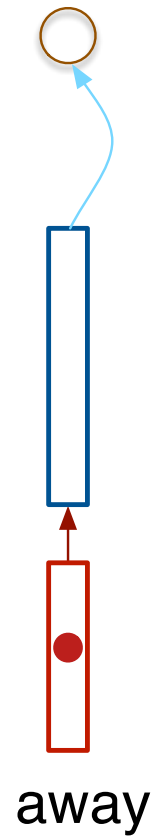
$\mathbf{h}_i$

$\mathbf{c}_i$

w_i



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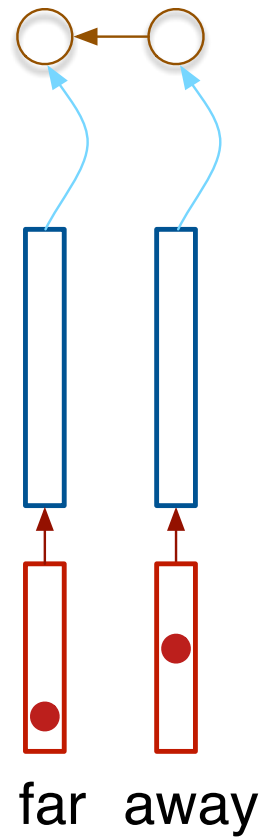

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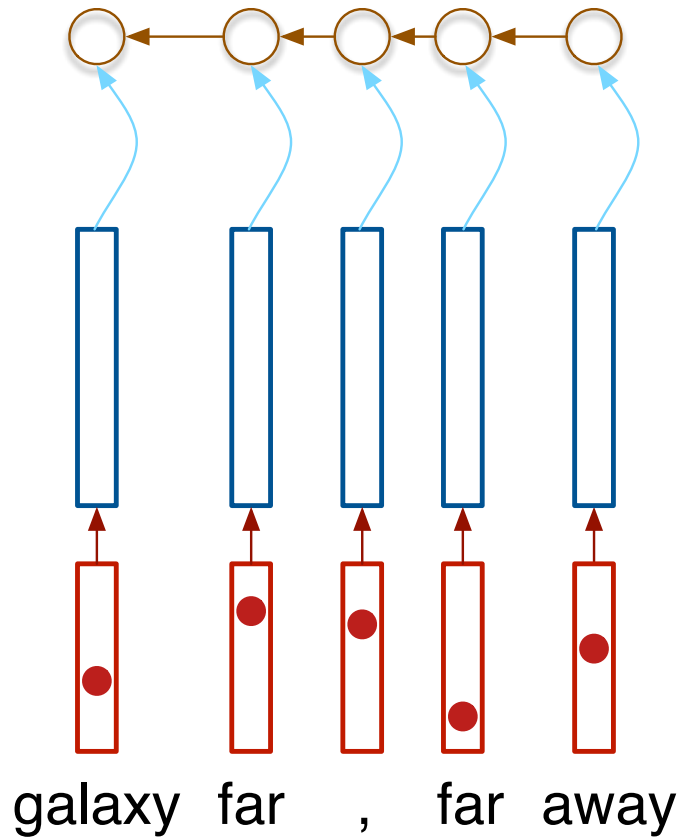

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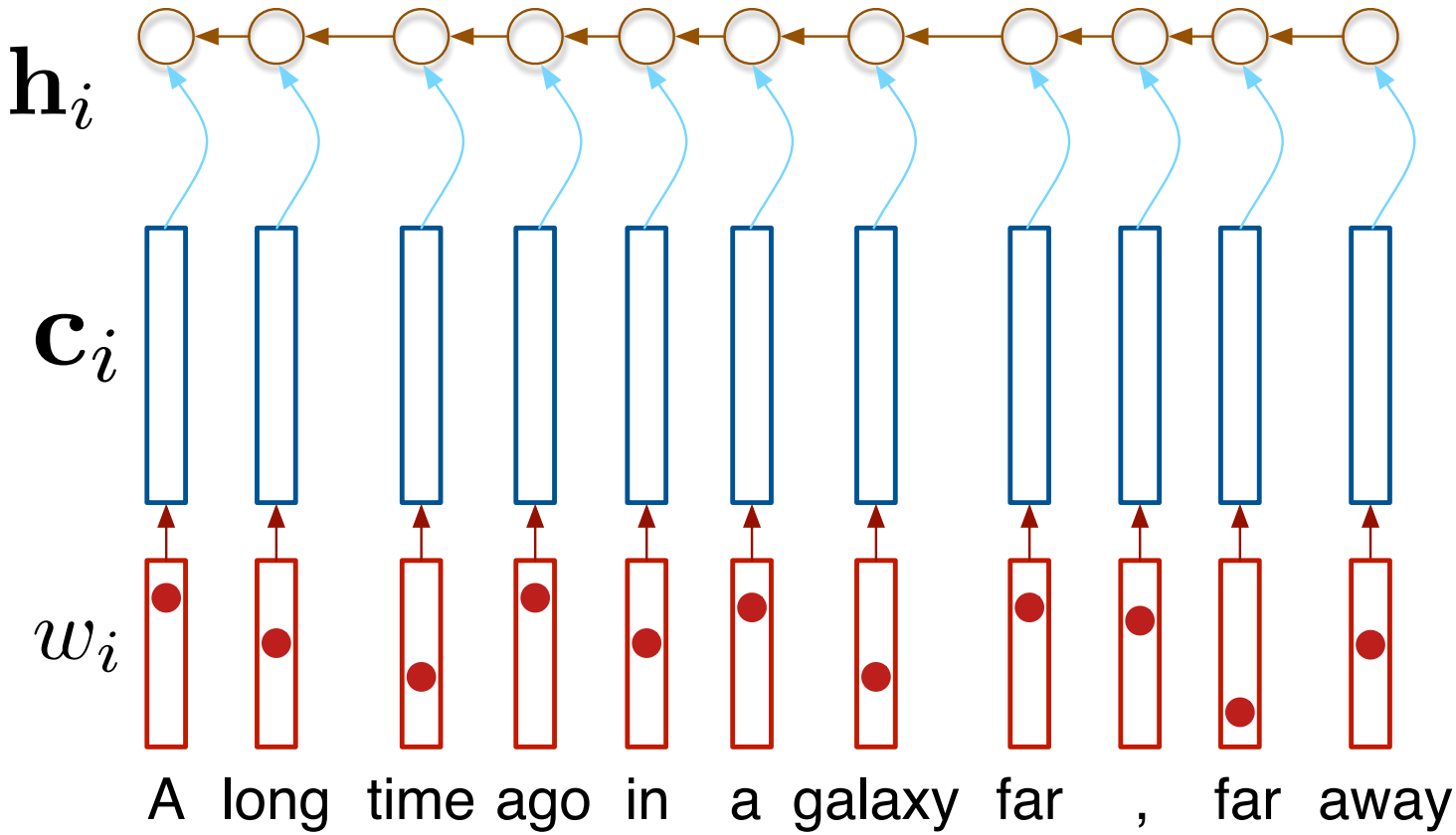

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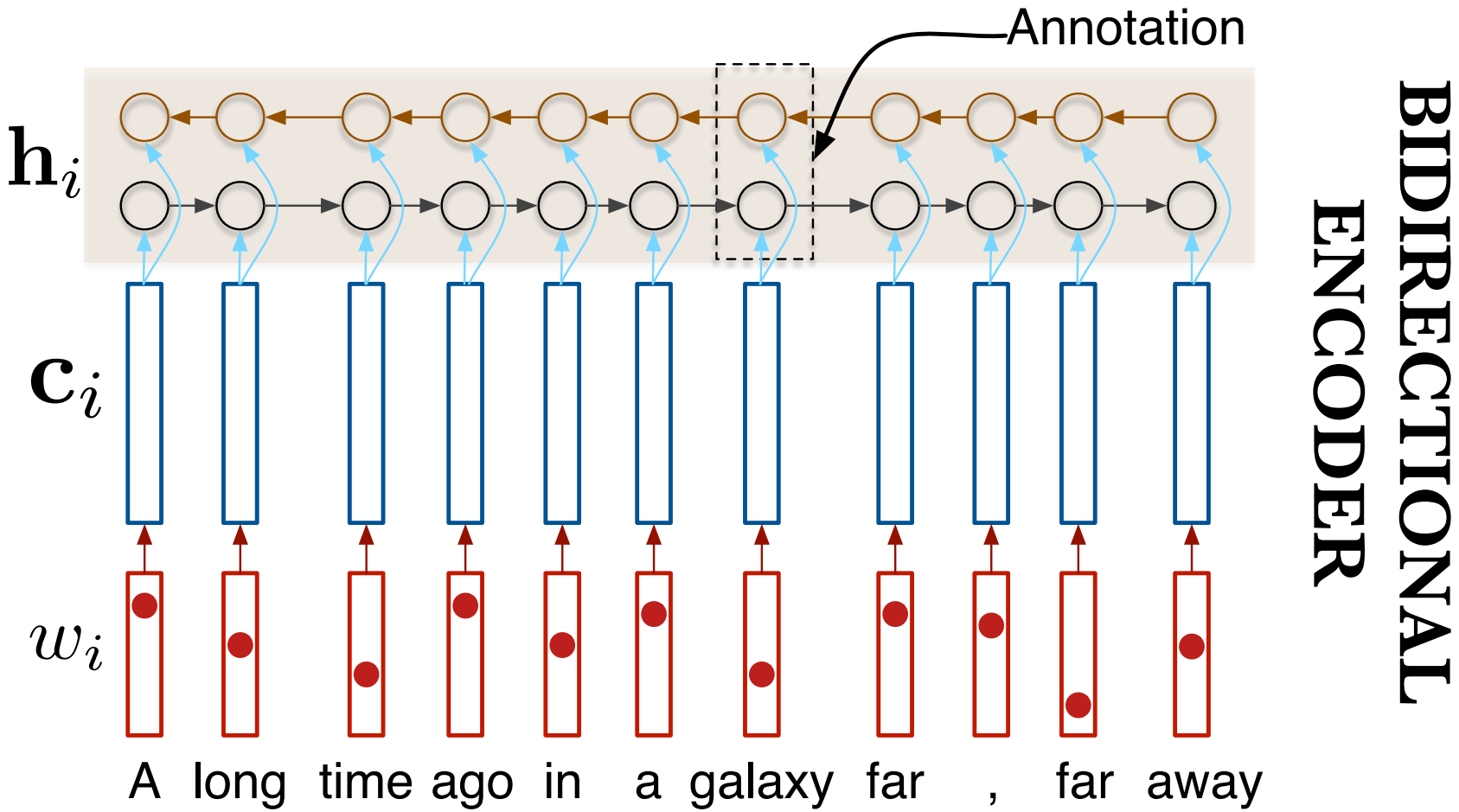
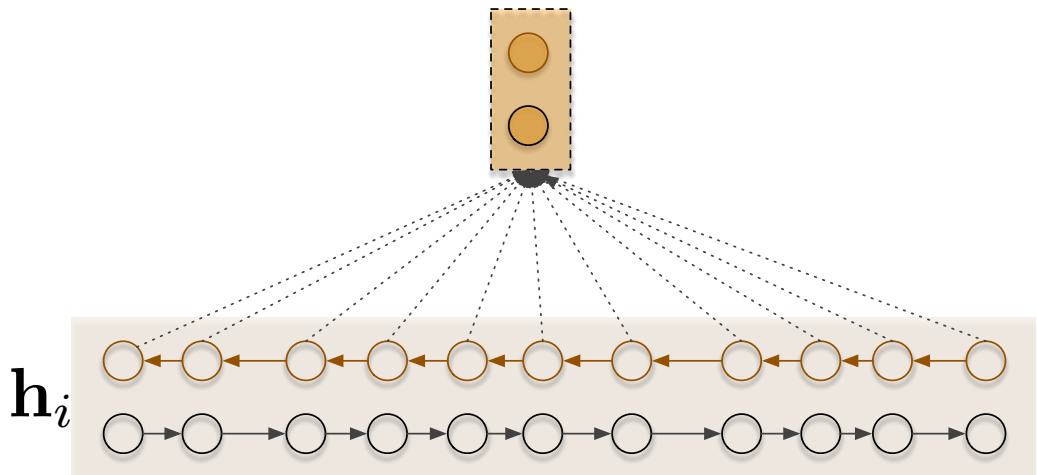


Pooling operation

Compute the feature-wise **average** or **maximum** **activation** of a set of vectors

Aim: sub-sampling $\rightarrow$ result is a vector!



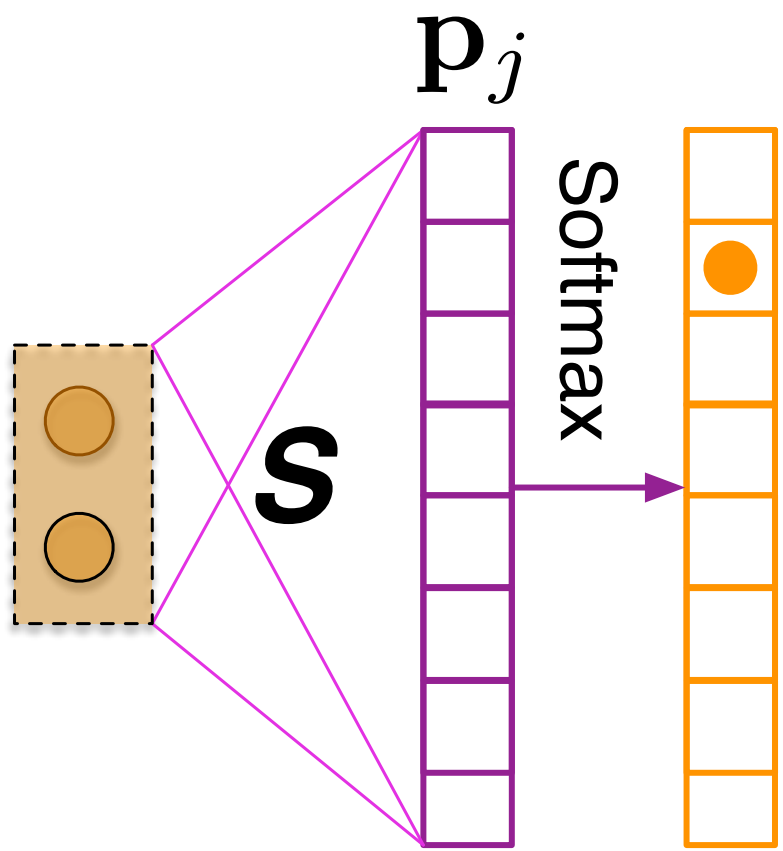
[Source: A comment on max pooling to read:

`https://mirror2image.wordpress.com/2014/11/11/geoffrey-hinton-on-max-pooling-reddit-ama/`]

Classification layer $\Rightarrow$ Softmax

Get a probability distribution by normalization $\rightarrow$ softmax:

$$p(\mathbf{c} = j|\theta) = \frac{e^{\mathbf{z}_j}}{\sum_{k=1}^{\|V\|} e^{\mathbf{z}_k}}$$



Text Processing: Deep Learning: Overview

- Shortest introduction to Neural networks
- Representing words
- Representing sentences
- **Deep Learning for Sentiment Analysis**
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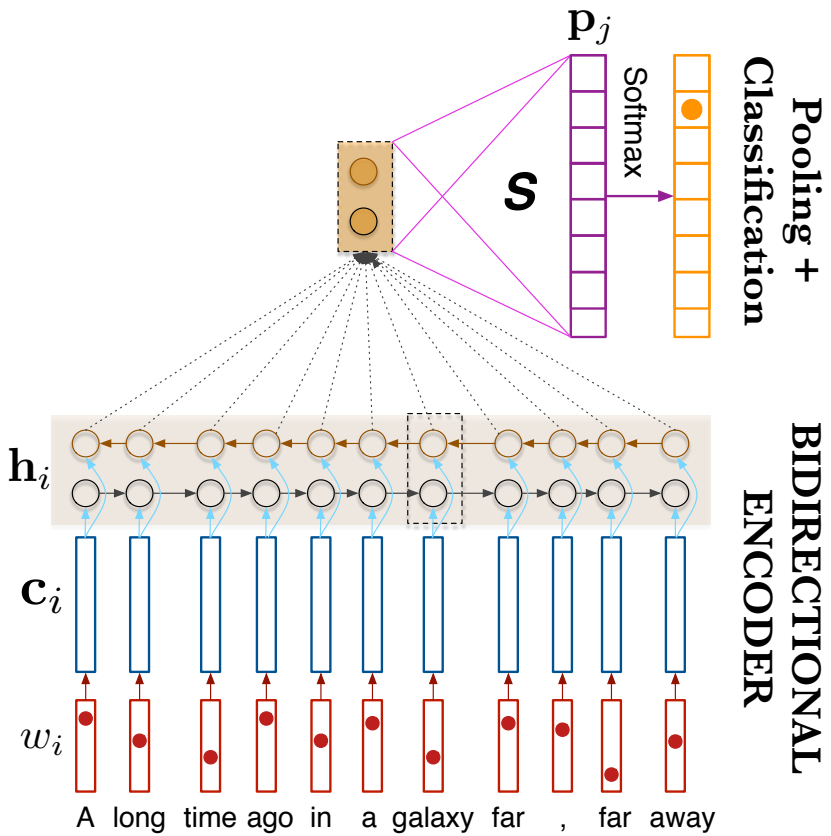
Deep Learning for Sentiment Analysis

Principle

Project or represent the **text** into a **continuous space** and train an estimator operating into this space to compute the probability of the sentiment.

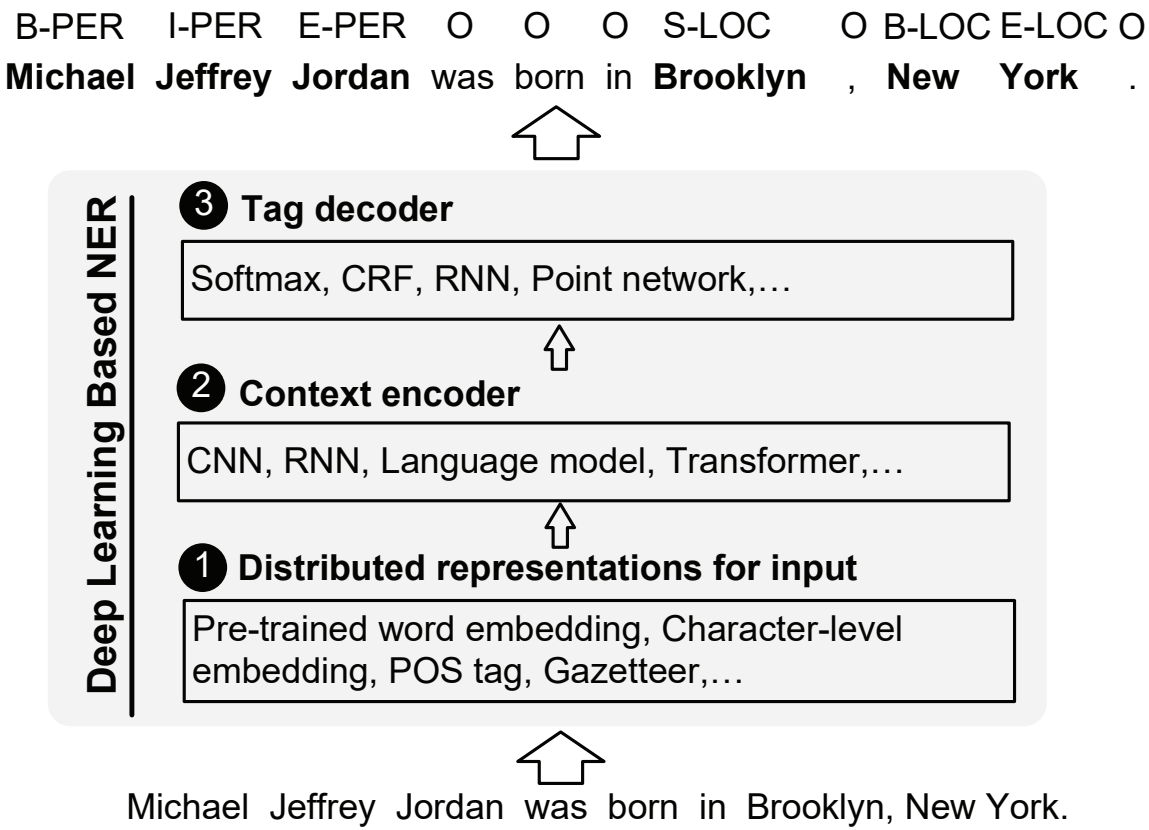


Deep Learning for SA

- Sahoo, C., Wankhade, M. & Singh, B.K. Sentiment analysis using deep learning techniques: a comprehensive review. Int J Multimed Info Retr 12, 41 (2023) https://doi.org/10.1007/s13735-023-00308-2 [Sahoo et al., 2023]

Deep Learning for NER

- Li, J., Sun, A., Han, J., & Li, C. (2018).
  A Survey on Deep Learning for Named Entity Recognition.
  `http://arxiv.org/abs/1812.09449` [Li et al., 2018]



Reading material

- Mikolov et al. (2013)
  Distributed Representations of Words and Phrases and their Compositionality.
  NeurIPS 2013.
  `https://arxiv.org/pdf/1310.4546.pdf`

What I couldn't talk about today

- **GloVe:** Global Vectors for Word Representation [Pennington et al., 2014]
- Recurrent cells addressing the problem of **vanishing gradient**
  - **LSTM:** Long Short-Term Memory [Hochreiter and Schmidhuber, 1997]
  - **GRU:** Gated Recurrent Unit [Cho et al., 2014]
- **Transformer models** → *Attention is All you Need* [Vaswani et al., 2017]
- **Contextualised word embeddings:**
  - **ELMo:** Embeddings from Language Models [Peters et al., 2018]
  - **BERT:** Bidirectional Encoder Representations from Transformers [Devlin et al., 2018]
  - **XLNet** [Yang et al., 2019]
- Large Language Models → **GPT** [Radford et al., 2018, Brown et al., 2020]

Text Processing: Deep Learning: Resources

- Deep Learning book:
  `https://www.deeplearningbook.org/`
  [Goodfellow et al., 2016]

- Set of notebooks for Sentiment Analysis in Pytorch
  `https://github.com/bentrevett/pytorch-sentiment-analysis`

- Enjoy the lectures COM4513/6513 Natural Language Processing - Spring 21-22

Credit: slides based on Dr Loïc Barrault's slides for the autumn 2020/2021 course.

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Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. (2017).

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